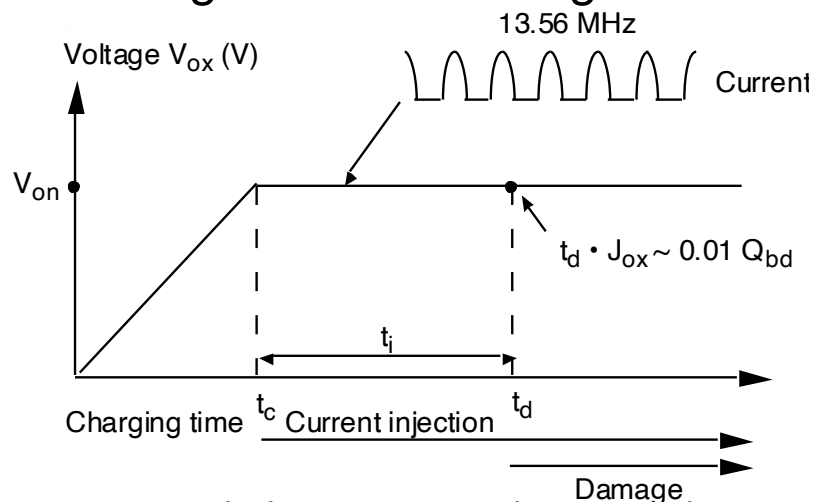


Plasma Process Induced Device Damage

電漿製程元件損害 11510CSR5419 Homework #02

1. Use realistic plasma/device parameters, prove plasma induced charging can cause gate oxide damage of antenna MOS structure within ms range



Typical plasma current/time number

J_i (mA/cm ²)	t_c (sec)	t_i (sec)	t_d (sec)
0.1	0.01	0.3	0.31
1	0.001	0.03	0.031
10	0.0001	0.003	0.0031

Please list all the details of assumptions including

1. Antenna MOS structure details and dimension
2. Plasma parameters and ion/electron currents
3. Charge to breakdown for such device
4. Etc.

List all the reference you quote, can not list AI as reference

Current injection on-set time

$$t_c = \frac{Q_{on}}{J_c \times A_f} = \frac{C_{total} \times V_{on}}{J_c \times A_f} = \frac{V_{on} \times \epsilon_{ox}}{J_c \times A_f} \left(\frac{A_f}{t_f} + \frac{A_g}{t_g} \right) = \frac{V_{on} \times \epsilon_{ox}}{J_c \times t_f} \left(1 + \frac{t_f}{t_g} \frac{A_g}{A_f} \right) \approx \frac{V_{on} \times \epsilon_{ox}}{J_c \times t_f}$$

Current injection to damage time

$$t_i = \frac{0.01 Q_{bd} \times A_g}{J_c \times A_f}$$

Total time to cause damage

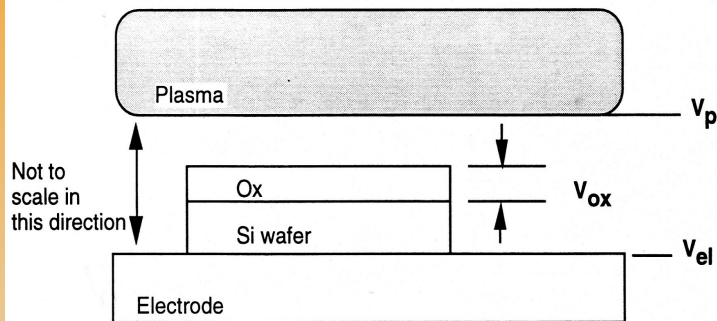
$$t_d = t_c + t_i = \frac{\frac{V_{on} \times \epsilon_{ox}}{t_f} + 0.01 Q_{bd} \frac{A_g}{A_f}}{J_c}$$

2. Read "CMOS plasma and Process Damage" Chapter 2, 9



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3. Charging is a problem in plasma etching. One cause for charging is a local imbalance in the ion and electron currents ($J_i \neq |J_e|$) to a wafer surface. In the modeling of charging it is assumed that the electrode voltage V_{el} and the properties of the plasma region, such as the potential V_p are the plasma density n_b (here given at sheath edge), are not affected by charging. (V_{el} , V_p and n_b are all constant)

- Assuming an excitation frequency of 13.56 Mhz and a sheath thickness of 0.5 cm. Explain using calculation why the rf voltage from the displacement current going across the sheath is not important in charging.
- Using the voltages as defined in the figure above and assuming that the silicon substrate is in good electrical contact with the electrode ($V_{Si} = V_{el}$), write down an expression that relates the charging current to the charging voltage V_{ox} that develops across the oxide. Neglect, the fact that in charging the voltages and currents can vary across the wafer surface.



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4. Difference between electron shading effect and plasma non-uniformity induced charging damage in metal etch: Since it is possible to have either plasma charging damage from non-uniform plasma or electron shading effect, design your test structure based from box and comb antenna structure that can identify the the difference. What should be the device parameter response difference for each case?

